Project Title

Multi-Pattern Search Engine Using Aho–Corasick Algorithm

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

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ABSTRACT

The Multi-Pattern Search Engine Using Aho–Corasick Algorithm is designed to efficiently search for multiple keywords simultaneously within a large text. The system uses a Trie data structure to store a collection of keywords and constructs failure links to improve the searching process. The Aho–Corasick algorithm allows the system to scan the input text in a single pass and identify all occurrences of the stored keywords efficiently. It reports each matched keyword along with its position in the text. The system supports Unicode characters and multilingual datasets, making it suitable for processing text in different languages. This approach significantly reduces the processing time compared with searching for each keyword separately. The proposed system can handle thousands of keywords and large text inputs effectively. It can be applied in search engines, intrusion detection systems, spam detection, log analysis, text filtering, and document processing, providing fast, scalable, and reliable multi-pattern matching.

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